

- 1 (a) A group of students investigated the effects of glucose and ATP on the contraction of muscle fibres.

Muscle tissue was cut into pieces along the length of the muscle. These pieces were pulled apart to obtain thin strips containing muscle fibres.

Fig. 1.1 shows the preparation of these thin muscle strips.

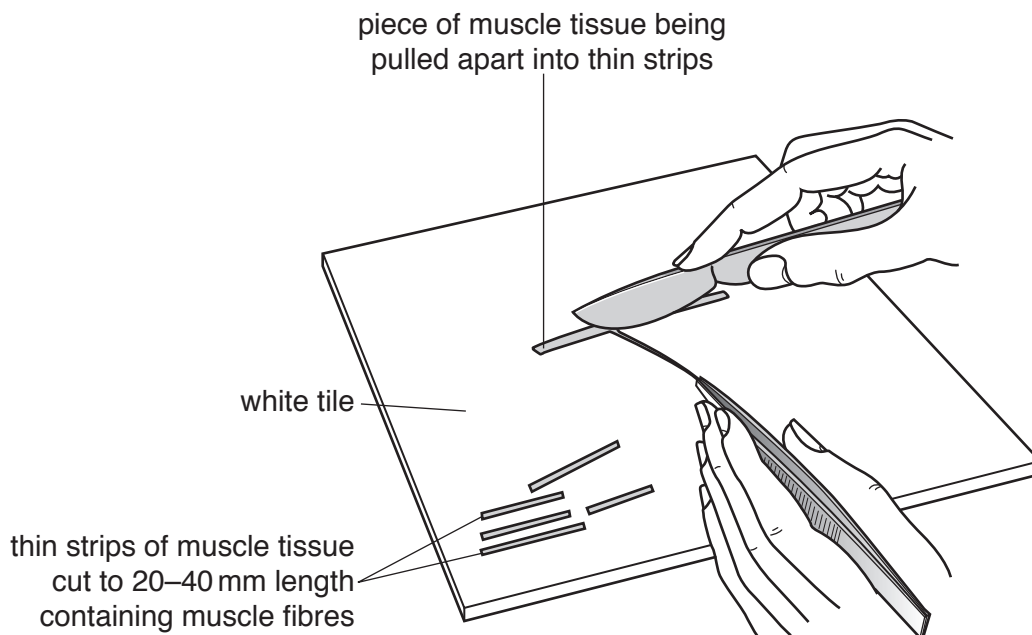


Fig. 1.1

- The prepared thin muscle strips were placed in a Petri dish containing Ringer's solution. Ringer's solution contains the correct balance of ions for animal tissues to function in the Petri dish.
- Some of these thin muscle strips were then placed on a microscope slide and their lengths were measured in mm.
- Ringer's solution was added to these thin muscle strips on the microscope slide and left for a few minutes.
- The length of each of the thin muscle strips was remeasured.

This procedure was repeated with two new sets of thin muscle strips. A different solution was added to each set of thin muscle strips on a microscope slide, instead of Ringer's solution, as follows:

- Ringer's solution containing glucose
- Ringer's solution containing ATP.

The percentage decrease in length was calculated for each of the thin muscle strips.

The results showed that the Ringer's solution containing ATP caused the thin muscle strips to decrease in length. Ringer's solution on its own and Ringer's solution containing glucose did not cause any change in the length of the thin muscle strips.

- (i) Identify the independent and dependent variables in the students' investigation.

independent variable

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dependent variable

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[2]

- (ii) State why the students calculated the decrease in length of the muscle fibres as a percentage.

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- (iii) State the purpose of testing the muscle strips in Ringer's solution without glucose or ATP.

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(b) The students decided to adapt their method to test the hypothesis:

the greater the concentration of ATP added to the thin muscle strips, the greater the percentage decrease in length.

The students were additionally provided with a stock solution of Ringer's solution containing 0.5% ATP.

Describe how the students could adapt their method to test this hypothesis.

Your method should be detailed enough for another person to follow.

You should not include details of preparing the thin muscle strips containing muscle fibres.

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- (c) The students' hypothesis was:

the greater the concentration of ATP added to the thin muscle strips, the greater the percentage decrease in length.

Complete Fig. 1.2 to show the graph predicted by this hypothesis. Add labels for the axes, including units, and draw the expected shape of the curve.



Fig. 1.2

[3]

- (d) The students carried out their planned method and plotted a graph of their results.

Suggest two reasons why the results obtained by the students may not be valid for the contraction of muscle fibres in a living organism.

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[2]

- (e) The students thought that adding glucose as well as ATP to muscle fibres might increase the effect of ATP.

Suggest how the students could find out if glucose has this effect.

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[Total: 18]